

EE 412/512 Project Part #2

Due December 4, 2025

In this part of the project, you will design a PCB for your amplifier.

The amplifier is to be built on an FR4 substrate with a thickness of 1 mm. The size of the PCB must be smaller than 2300 mils $\times$ 1000 mils. You will make patterns only on the top side of the PCB. The bottom side will be completely covered with a copper ground plane. Place the input and output SMA connectors on the short edges of the rectangular PCB. It is desirable that the elements nearest to the transistor on both sides are transmission lines for ease of mounting the transistor.

First, modify your circuit to use the available surface-mount components. Replace the generic transmission lines in your design with microstrip lines and determine their dimensions. You may want to adjust the topology of your circuit to make it easier to tune later on. You can use open-circuited low-impedance transmission lines for small-value shunt capacitors. You can cut (or add) copper later to reduce (or to increase) the capacitance in the testing phase. You can use short-length high-impedance transmission lines to generate series inductors.

Carefully set up your schematic so that the main part of the layout is automatically generated. You can insert surface-mount components using the proper element, specifying the component size. You need to modify the layout manually to insert the necessary ground pads on the top layer. You need to insert via holes to make ground connections with a minimum of inductance. This is especially important for the emitter-to-ground connection. Do not put too many via holes; keep them fewer than 80 holes. You need to manually put wires through these holes!

Your schematic should correctly correspond to your PCB. Make a simulation of the updated schematic after PCB design.

There should be pads for the supply connection on the top surface.

In your report, provide

1. A full schematic of your design, including the biasing network.
2. The simulation results show that the specs are satisfied.
3. An image of your PCB design in your report.
4. An image of your Gerber file of the top layer.
5. An image of the NC Drill file showing the drill locations.

Generate the top copper Gerber file, the bottom copper Gerber file (to specify the PCB dimensions), and a drill file that specifies drill locations. Place them in a zip file for uploading to Moodle. Before uploading it, inspect it in

<https://www.pcbway.com/project/OnlineGerberViewer.html>

Your PCB will be manufactured after your design is inspected by the same tool and approved.

You will lose points if you make mistakes:

- Unconnected components on the PCB
- Shorted components on the PCB
- Missing component footprints
- Overlapping (or outside the board limits) transmission lines or components
- DC paths to ground, destroying the DC bias.
- PCB larger than the limit
- Gerber and/or drill files not correctly generated
- Components too close to SMA connectors
- Very small ground pads
- Missing vias for ground pads
- SMA connectors not at opposite ends of PCB.
- Transmission lines leading to SMA connectors are not 50Ω
- No decoupling capacitors at the bias circuit or supply.